PRODUCT WHITEPAPER

Three-Phase Padmount Transformer

Engineering reference for underground distribution systems — design features, standard ratings, construction, and factory testing for Forlantis three-phase pad-mounted transformers.

45–7,500
KVA CAPACITY RANGE

40.5 kV
MAX PRIMARY VOLTAGE

<8 wks
DISTRIBUTION-READY DELIVERY

Overview

The Forlantis three-phase padmount transformer is designed for use in underground distribution systems. Sealed high-voltage and low-voltage safety compartments ensure safe operation and reduce the risk of accidents, making these units well suited for residential developments, commercial sites, hospitality, and institutional buildings. Units are housed in a lockable, door-mounted cabinet and are typically installed outdoors on a concrete pad.

Two basic configurations are available — radial and loop feed — selected based on the distribution circuit the transformer will serve. Windings are available in aluminum or copper, optimized to balance efficiency against footprint. All units are built to the latest applicable ANSI, DOE, IEEE, CSA, and IEC standards.

Design Features

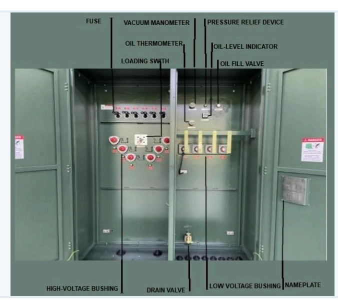
- Mild steel tank, optional stainless steel
- Loop feed or radial feed configuration
- Insulation fluid: mineral oil or FR3 natural ester (vegetable oil)
- High operating efficiency
- Low sound level
- 50/60 Hz design

Standard Particulars

- Three-point latching door for security
- Removable sill for easy installation
- Stainless steel cabinet hinges and mounting studs
- Steel HV/LV compartment barrier
- Penta-head padlocking on HV and LV doors
- Five-legged core/coil assembly
- Self-actuating pressure relief valve
- Nameplate per ANSI requirement

Three-phase unit — radiator banks extended

Cabinet doors closed



HV/LV compartment interior

Specifications

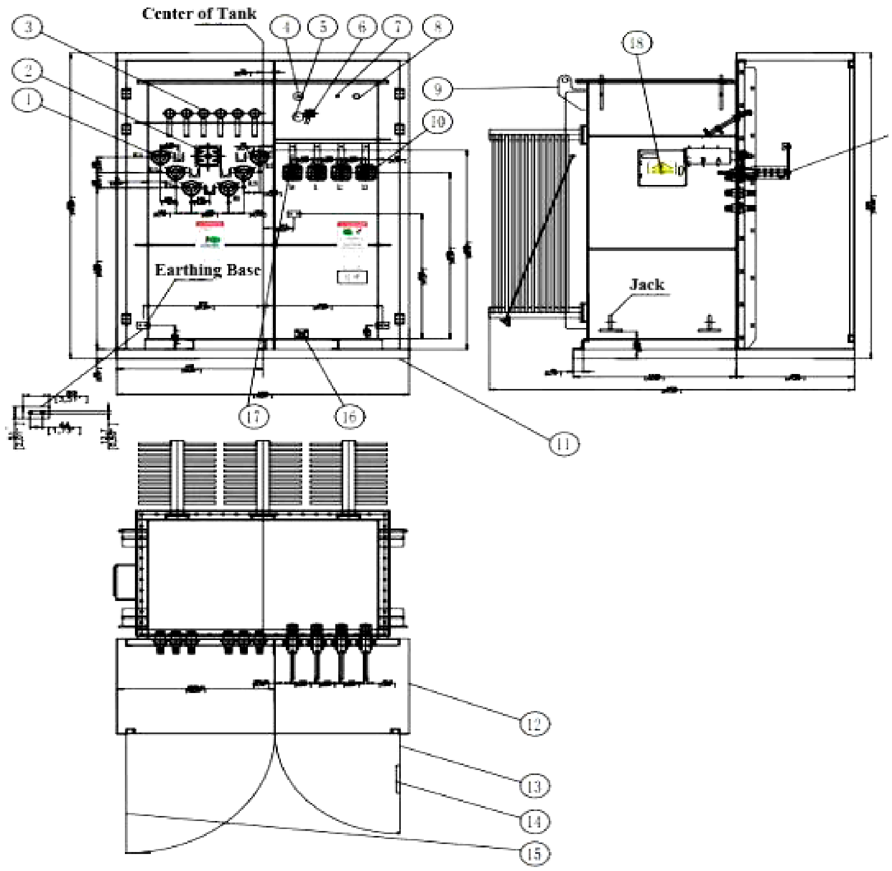
Capacity range	45 – 7,500 kVA
Primary voltage	40.5 kV and below
Secondary voltage	120/240V – 480V
Cooling	ONAN, 50/60 Hz design
Insulating fluid	Mineral oil or FR3 natural ester (vegetable oil)
Tank construction	Mild steel; stainless steel optional
Feed configuration	Radial or loop feed
Standards	UL, NEMA, ANSI/IEEE, DOE, CSA, IEC

Standard Ratings & Dimensions

CAPACITY (KVA)	NO-LOAD LOSS (W)	LOAD LOSS (W)	W (MM)	D (MM)	H (MM)	OIL VOL. (L)	WEIGHT (KG)
45	160	1000	1730	990	1270	416	950
75	180	1250	1730	990	1270	435	1020
112.5	200	1500	1730	1245	1270	454	1070
225	400	3050	1830	1295	1270	530	1430
300	480	3650	1830	1295	1270	605	1655
500	680	5100	2260	1345	1270	720	2110
750	980	7500	2260	1448	1625	1022	2950
1000	1150	10300	2260	1500	1625	1325	3720
1500	1640	14500	2260	2190	1854	1552	4672
2500	2680	27786	1830	2515	1854	2006	6580

Other ratings not listed above can be custom-designed to meet specific project requirements.

Loop Feed Configuration

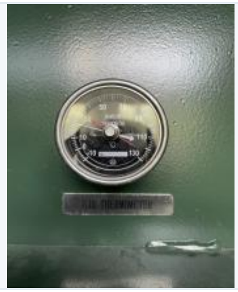


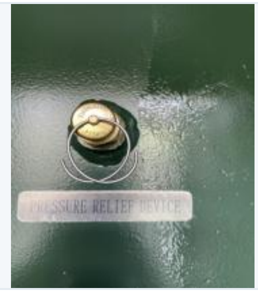
Optional Accessories

- Oil level gauge, liquid temperature gauge, pressure vacuum gauge
- Welded cover with hand hole
- Oil drain valve with or without sampler
- Mechanical pressure relief device on tank cover
- Primary switching: LBOR oil switch, de-energized tap changer, dual voltage switch, sectionalizing loop switches
- Over-voltage protection: distribution-class metal oxide or valve-type arresters
- Over-current protection: bayonet expulsion fuses, weak-link cartridge fuses, current-limiting fuses
- CTs/PTs with mounting support; externally mounted kWh meter
- Stainless steel tank/cabinet, or partial stainless design
- Flip-top low-profile cabinet; hinged weather cover; custom paint

Accessory Detail

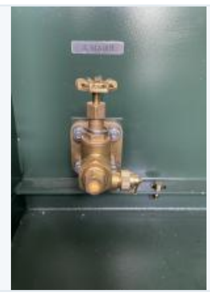

Vacuum manometer


Oil thermometer


Pressure relief device


Oil-level indicator


Oil fill valve


Oil drain valve

Construction

Core

Three-legged, step-lap mitered cores are manufactured with precision cutting technology and stacked to minimize corner joint gaps and maximize efficiency. For wye–wye connected units and other specialized applications, five-legged wound cores or shell-type triplex designs are used. Cores are built from grain-oriented, precision-cut, burr-free silicon steel, with multiple steel grades available to optimize core loss for the application.

Coil

Coils use a rectangular configuration with wire-wound high-voltage primaries and sheet-wound secondaries, minimizing axial stress during short-circuit events and maintaining magnetic balance across tap connections. Coils are wound on precision machines with strict tension control, then insulated with diamond-pattern epoxy-coated paper — reinforced at high-stress points — and heat-cured under calculated hydraulic pressure for long-term reliability.

Active Part

The core and coil assembly is braced with heavy-duty steel end frames to prevent distortion of the rectangular coils under short-circuit conditions, and clamped using precision presses for a rigid, durable structure. Each active part is engineered to exceed ANSI and IEEE short-circuit performance requirements, with impedance shift after a fault comparable to circular-wound coil designs.

Tank

Tanks are fabricated from precision-cut, hot-rolled, pickled, and oiled steel — fully welded and hermetically sealed to protect the insulating fluid and internal components. The finishing system exceeds IEEE Std C57.12.28-2005: an eight-stage pretreatment process converts the surface to a nonmetallic, corrosion-resistant iron phosphate layer, followed by a dual-layer epoxy primer (E-coat) and two-component urethane topcoat for long-term UV and corrosion resistance.

> **Vacuum processing.** Transformers are dried and filled with filtered insulating fluid under vacuum, with secondary windings energized during the process to remove residual moisture and ensure maximum fluid penetration into the insulation system for long-term dielectric strength.

Insulating Fluid

Units are available with either electrical-grade mineral insulating oil or Envirotemp FR3 natural ester fluid. Both are tested, degassed, and chemically stabilized to minimize acid ions and oxidation, then re-tested for dryness and dielectric strength before being vacuum-filled into the completed transformer. FR3 natural ester fluid is bio-based, biodegradable, and non-toxic per acute aquatic and oral toxicity standards, offering extended insulation life and superior fire safety.

Routine Factory Tests

- **Ratio, polarity & phase relation** — winding ratios, tap voltages, insulation clearances
- **Resistance test** — HV/LV connection integrity, loss calculation data
- **Applied potential test** — dielectric strength of HV/LV windings to ground
- **Induced potential test** — 3.46× rated voltage + 1000V for reduced neutral designs
- **Loss tests** — no-load loss, excitation current, impedance voltage, load loss
- **Leak test** — tank pressurized to 7 psig
- **Device operation test** — auto/manual sequencing of all electro-mechanical devices

Design Performance Tests

- **Temperature rise test** — automated heat run to ANSI/IEEE limits
- **Audible sound test** — confirms NEMA sound level requirements
- **Lightning impulse test** — reduced, chopped, and full wave sequence
- **Optional tests** — short-circuit withstand, lifting/handling verification, and other customer-specified tests

Ready to spec this equipment? Contact our solutions engineers for ratings, lead times, and delivery at **info@forlantisgrid.com** or visit **forlantisgrid.com/contact**.